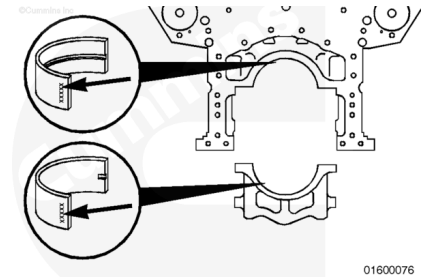


Trainings ▾

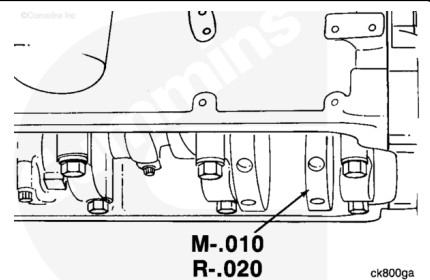
General Information

The upper bearings contain an oil hole. The lower bearings do **not** have an oil hole. Both bearings are marked on the backside indicating location (upper or lower) and size (standard or oversize). The amount that the bearing is oversize is indicated in U.S. customary inches.

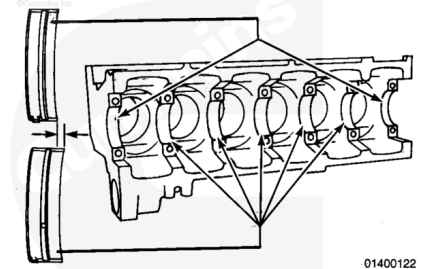
Use the same size bearing, standard 0.254 mm [0.010 in], 0.508 mm [0.020 in], or 0.762 mm [0.030 in] that was removed.



The crankshaft will be stamped on the end of the number 1 counterweight to indicate whether it has been ground undersize. The amount of material removed from the main journals and the rod journals are both stamped at this location. Thrust bearing size is stamped on a crankshaft counterweight adjacent to the thrust location.



The location numbers of the main bearing caps, beginning at the front, are number one through number 7 on QSK45 engines, and number 1 through number 9 on QSK60 engines.



Preparatory Steps

⚠ WARNING ⚠

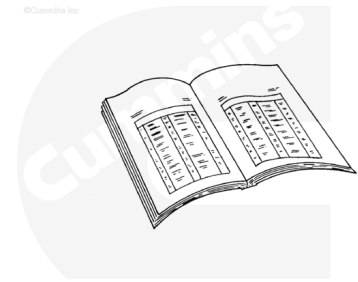
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged

contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html>)
- Remove the oil suction and supply tubes and brackets. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html>)
- Remove the oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html>)



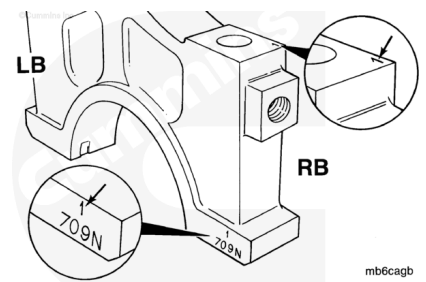
ck800wa

Remove

⚠ WARNING ⚠

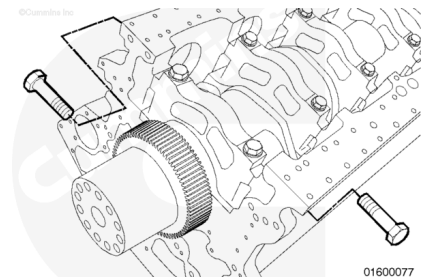
Replace the main bearings one at a time. Personal injury and damage to the crankshaft can result if the crankshaft falls.

The main bearing caps **must** be marked according to their respective positions (number 1 through number 7 for QSK45 engines and number 1 through number 9 for QSK60 engines with number 1 at the front of the engine). Each cap **must** also have the block number stamped into it. Use a steel stamp to mark any cap **not** correctly marked.



Note : Access to some side capscrews involves the removal of other components.

Remove the two side capscrews.



Remove the main bearing cap mounting capscrews.

Remove the cap. Use the main bearing cap puller, Part Number 4919357.

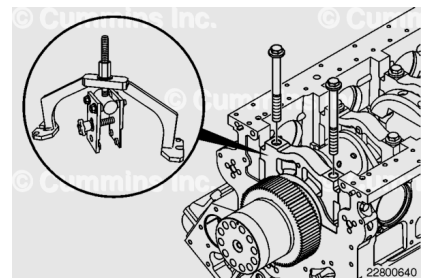
Align the main bearing cap puller with the main bearing cap.

Install the fixture to the left and right oil pan rails of the engine block.

Install the main bearing cap puller clamp plate onto the main bearing cap and tighten the clamp knob.

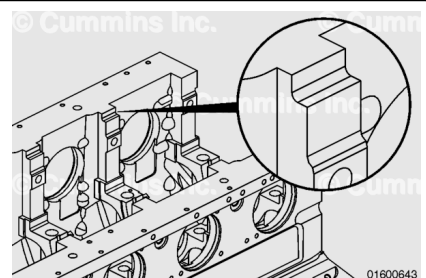
Turn the adjusting nut at the top of the fixture with a wrench to lift the main bearing cap.

Note : If the flywheel housing is still in place, the rear main bearing cap will need to be tilted forward slightly for tool access. Use a pry bar or a large screwdriver to tilt the cap forward.



QSK60 engines which do **not** have the machined ledge on the left bank pan rail inner wall, as shown in the illustration, and have a fully counterweighted crankshaft installed, **must** have the number 14 and 15 crankshaft counterweights removed in order to remove the number 8 main bearing cap.

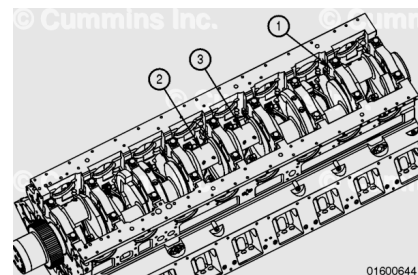
Use the following procedure for the removal and installation of the crankshaft counterweights. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-016-tr.html>)



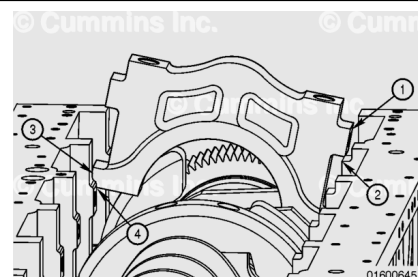
Note : QSK60 cylinder blocks which do not have the machined ledge can have this feature added. Use the following procedure Machine section for the pan rail machining instructions. Refer to Procedure 001-026 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-026.html>)

QSK60 engines which do have the machined ledge to the left bank pan rail inner wall and have a fully counterweighted crankshaft installed on the number 8 main bearing cap can be removed without removing any counterweights.

Note : For removal of the number 8 main bearing cap, with a fully counterweighted crankshaft installed, the number 4 and 5 connecting rod journals must be positioned at bottom dead center (BDC).



With the main bearing cap clear of the interference pads, extract using service tool, Part Number 4919357, to lift the main bearing cap until the left bank upper interference pad of the main bearing cap (1) passes the cylinder block machined ledge (2). Angle the cap so that the right bank lower interference pad of the main bearing cap (3) passes the upper interference pad of the cylinder block (4).

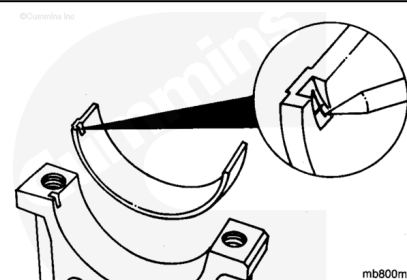


Remove the main bearing cap.

Note : The main bearing capscrews must be removed during the process.

Note : All main bearing cap locations which have the left bank pan rail inner wall machined ledge can have the main bearing cap removed using this method.

Remove the bearing from the cap, and mark the location in the tang for future identification or analysis.



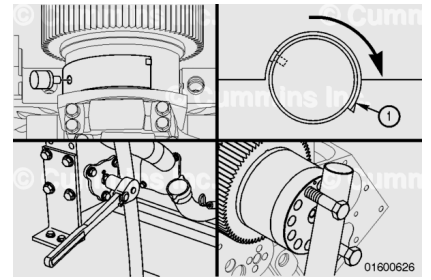
Note : If the crankshaft is being removed, the upper main bearing shells do not require removal at this time.

To remove the upper main bearing shells, use the main bearing remover, Part Number 3823818.

Rotate the crankshaft so the tang (1) of the main bearing shell rolls out of the block first.

Use the barring device to slowly rotate the crankshaft.
Continue rotating the crankshaft until the bearing shell is out of the engine block. If the flywheel housing has been removed, the crankshaft can be turned using the crankshaft adapter mounting capscrews.

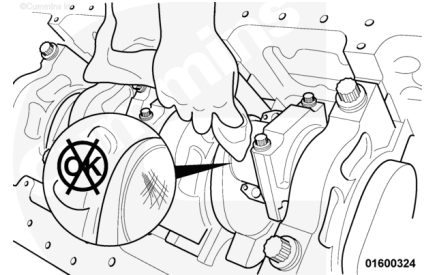
Mark the upper main bearing location in the tang area of the bearing.



Note : If bearing replacement was initiated due to damage, check the oil pan for debris. If significant debris is present, remove and disassemble the lubricating oil pump and inspect the bushing for debris damage. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html>)

Clean the crankshaft journal with a lint-free cloth.

Inspect the journal for damage.



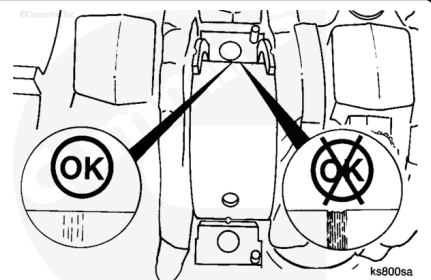
Clean and Inspect for Reuse

⚠ CAUTION ⚠

To reduce the possibility of damage to the crankshaft, do not use a scraper or a wire brush on the crankshaft journals.

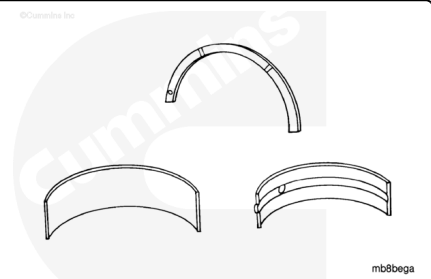
Use a lint-free cloth to clean the crankshaft journal.

Inspect the crankshaft journal for damage.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

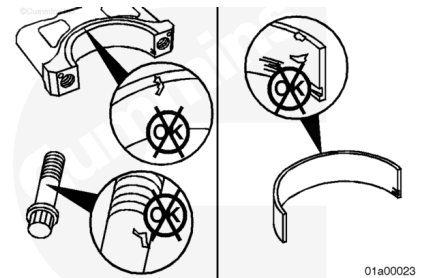
To reduce the possibility of bearing damage, do not use a scraper or wire brush on bearing surfaces.

Clean the bearings with solvent and dry with compressed air.

Inspect the main caps, capscrews, and washers for the following damage:

- Pitting
- Cracking
- Thread damage.

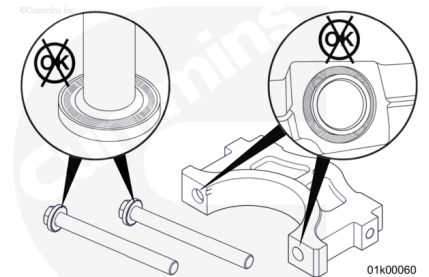
Replace any damaged part.



Inspect the main caps, capscrews, and washers for the following damage:

- Galling under capscrew head or around capscrew hole in the main bearing cap.

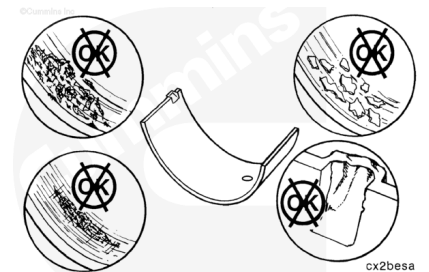
Replace any damaged part.



Inspect the bearings for damage.

Replace any bearings that have the following damage:

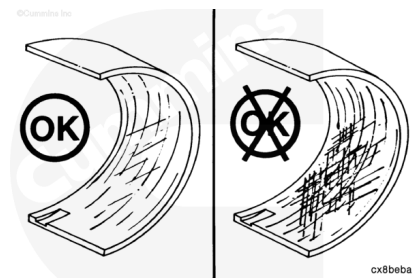
- Nicks
- Cracks
- Burrs
- Scratches (deep enough to be felt with a fingernail)
- Scrapes
- Gouges
- Embedded particles
- Fretting
- Pitting
- Flaking
- Corrosion
- Lock tang damage.



Normal bearing wear produces a smooth finish, which will wear into the copper lining. Exposed copper does **not always** indicate worn bearings. Reference the Parts Reuse Guidelines, Bulletin 3810303.

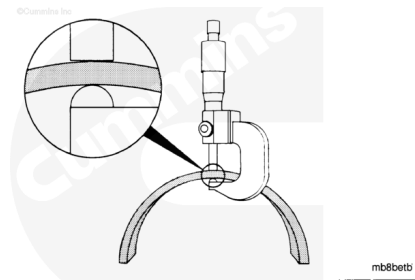
If large areas of copper lining are visible in the bearings before the engine has accumulated 3000 hours, inspect the engine for contamination from fine dirt particles and correct the problem.

For more detailed information of bearing damage, reference the Analysis and Prevention of Bearing Failures Manual, Bulletin 3810387.



⚠ CAUTION ⚠

To reduce the possibility of damage to the bearing surfaces, use adequate professional tools and measuring techniques. Use of improper measuring tools or methods may cause local surface damages that can become the initiation point for bearing surface layer failure.



Use a ball-end micrometer to measure the bearing thickness at the wear location.

Main Bearing Thickness			
	mm		in
Standard	4.317	MIN	0.1699
	4.337	MAX	0.1707
Oversize 0.25 mm [0.010 in]	4.442	MIN	0.1748
	4.462	MAX	0.1756
Oversize 0.50 mm [0.020 in]	4.567	MIN	0.1798
	4.587	MAX	0.1805
Oversize 0.75 mm [0.030 in]	4.692	MIN	0.1847
	4.712	MAX	0.1855
Oversize 1.0 mm [0.039 in]	4.817	MIN	0.1896

Main Bearing Thickness

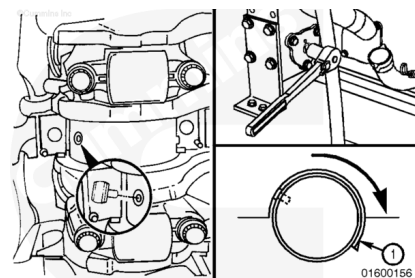
	mm		in
	4.837	MAX	0.1904

Install

Used bearings **must** be installed in their original location.

Install the upper main bearing shells using the same method that was used for the removal of the shells.

The bearing tang **must** fit into the slot in the bearing saddle to secure the bearing in the proper location.

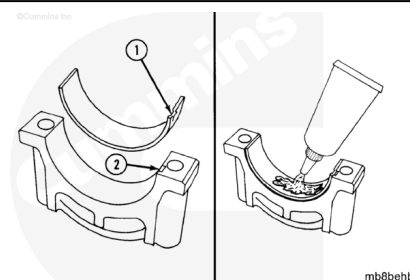


Use a lint-free cloth to clean the lower main bearing cap, lower thrust bearings, and mounting surfaces.

Do **not** lubricate the backs of the main bearings.

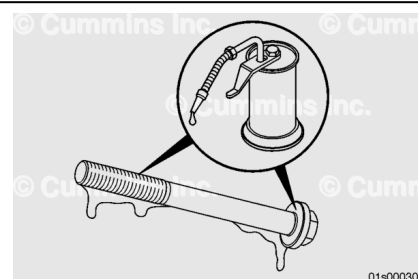
Align the tang (1) in the bearing with the slot (2) in the main bearing cap. Install the bearing. The end of the bearing **must** be even with the main bearing cap mounting surface.

Lubricate the bearing surface with Lubriplate™ 105, Part Number 3163086.



⚠ CAUTION ⚠

To reduce the possibility of engine damage, the grooves in the thrust bearings must point toward the crankshaft. The dowels that secure the bearings must not protrude above the bearing.

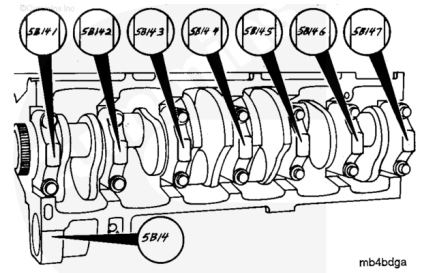


Lubricate the capscrew head and threads with Society of Automotive Engineers (SAE) EP 140W oil. Allow the excess oil to drip off the capscrews before installing them in the block.

Install the two thrust bearings on the number 6 main bearing cap on QSK45 engines and the number 8 bearing cap on QSK60 engines. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-007-tr.html>)

⚠ CAUTION ⚠

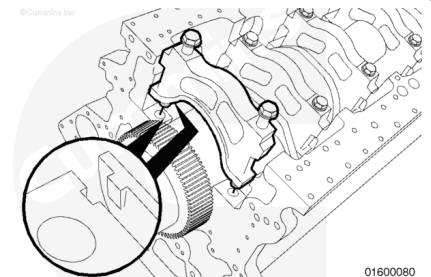
To reduce the possibility of engine damage, the numbers on the main bearing caps must be the same as the numbers on the block.



Check the number on the main bearing caps. The last digit (1 to 9) of each number indicates the correct location.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the side of the cap with the bearing tang slot and the bearing locking tang are toward the same side of the block.



Place the ring dowels in the capscrew holes.

The bearing shells **must** be firmly seated in the cap and the correct capscrew in position. Install the main bearing caps.

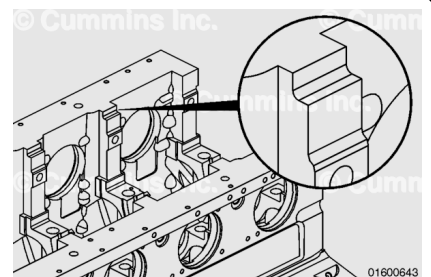
Do **not** hit the main bearing caps with a hammer. The bearing shells can fall out or be dislodged.

Keep the cap straight while moving it past the side capscrew holes.

QSK60 engines which do **not** have the machined ledge to the left bank pan rail inner wall, as shown in the illustration, and have a fully counterweighted crankshaft installed, **must** have the number 14 and 15 crankshaft counterweights removed in order to install the number 8 main bearing cap.

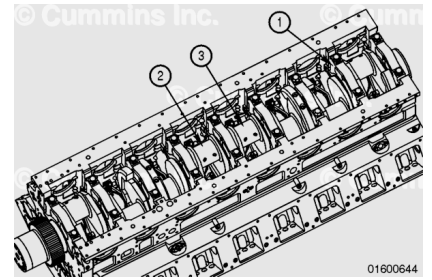
Use the following procedure for removal and installation of the crankshaft counterweights. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-016-tr.html>)

Note : QSK60 cylinder blocks which do **not** have the machined ledge can have this feature added. Use the following procedure Machine section for machining the pan rail machining instructions. Refer to Procedure 001-026 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-026.html>)

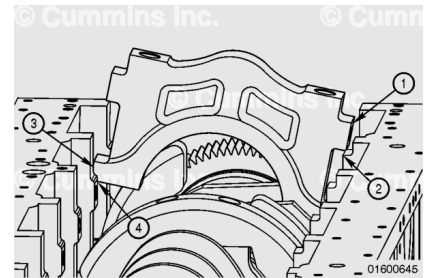


QSK60 engines which do have the machined ledge to the left bank pan rail inner wall, and have a fully counterweighted crankshaft installed, can have the number 8 main bearing cap installed without removing any counterweights.

Note : For removal of the number 8 main bearing cap with a fully counterweighted crankshaft installed, the number 4 and 5 connecting rod journals must be positioned at BDC.



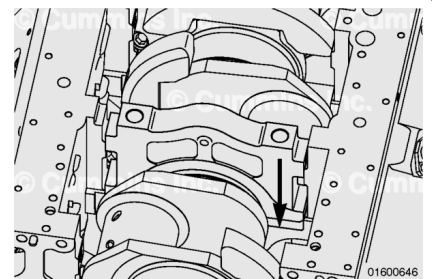
Angle the main bearing cap so the left bank cap side is lower than the right bank cap side. Allow the left bank upper interference pad of the main bearing cap (1) to sit in to the machined ledge (2). Carefully lower the right bank side of the main bearing cap so the right bank lower interference pad of the main bearing cap (3) passes the upper interference pad of the cylinder block (4).



Note : The main bearing capscrews must be removed during the process.

Lower the cap into place.

Continue to install the main bearing caps as in previous steps.

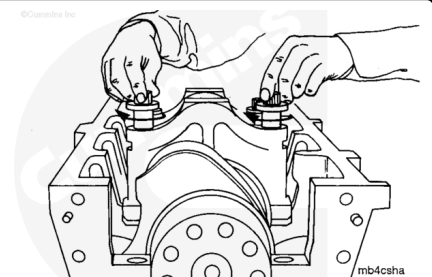


⚠ CAUTION ⚠

Do not rotate the crankshaft until all of the main bearing caps are pulled to the block. Damage will result if the bearings move out of location.

Install the capscrews by hand.

Turn the two capscrews by hand at the same time until each touches the main bearing caps.

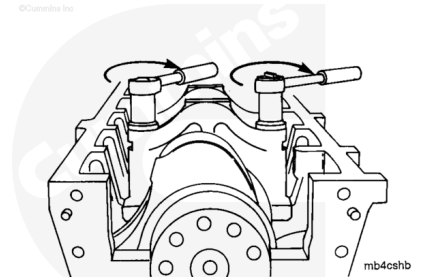


Do **not** use an impact wrench. The main bearing shells can become dislodged.

Use two wrenches to turn both capscrews on a main bearing cap at the same time to seat the cap into position.

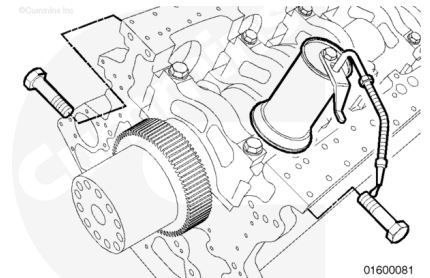
Seat all of the main bearing caps before tightening.

Check to make sure the number 7 cap on QSK45 engines (the number 9 cap on QSK60 engines) is even with the back of the block.



Lubricate the side capscrews with clean oil, allowing the excess to run off.

Turn the capscrews in by hand as far as possible.



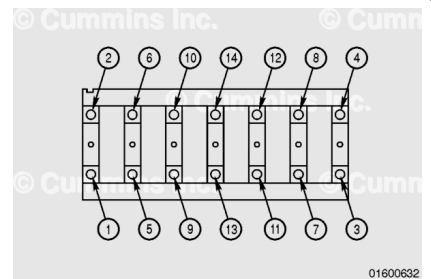
QSK45 Engines

Use the following steps to tighten the capscrews in the sequence shown.

Tighten the main bearing cap containing the thrust bearings to 50 N•m [37 ft-lb] before tightening to specification.

Torque Value:

1. 195 N•m [144 ft-lb]
2. 420 N•m [310 ft-lb]
3. Loosen all
4. 200 N•m [148 ft-lb]
5. 420 N•m [310 ft-lb]
6. Rotate 90 degrees



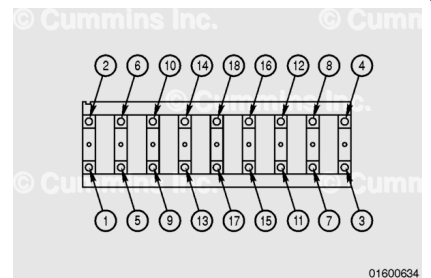
QSK60 Engines

Tighten the capscrews in the sequence shown.

Tighten the main bearing cap containing the thrust bearings to 50 N•m [37 ft-lb] before tightening to specification.

Torque Value:

1. 195 N•m [144 ft-lb]
2. 420 N•m [310 ft-lb]
3. Loosen all
4. 200 N•m [148 ft-lb]
5. 420 N•m [310 ft-lb]
6. Rotate 90 degrees



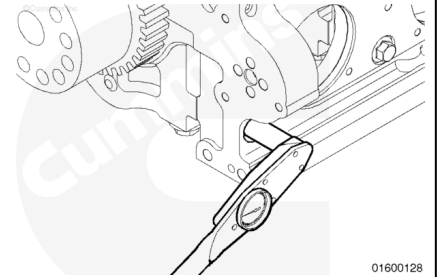
QSK45 and QSK60

Tighten the right bank main bearing cap side capscrews first.

Torque Value: 305 n•m [225 ft-lb]

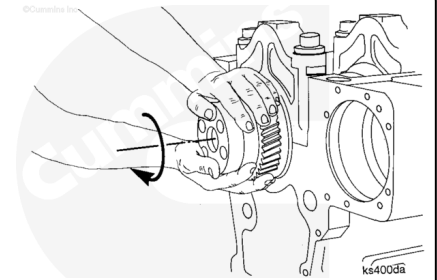
Tighten the left bank main bearing cap side capscrews after the right bank.

Torque Value: 305 n•m [225 ft-lb]

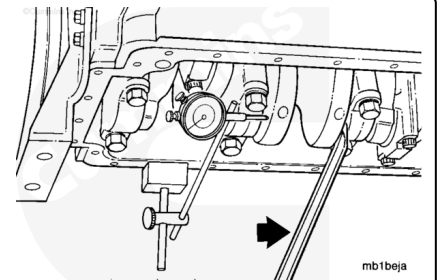


Turn the crankshaft with the barring mechanism.

Note : The crankshaft will turn freely if the main bearings are installed correctly.



Check the crankshaft end clearance. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-007-tr.html>)



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html>)

- Install the oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html>)
- Install the oil suction and supply tubes and brackets. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html>)
- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Start and operate the engine to 70°C [158°F] coolant temperature.
- Inspect for coolant and oil leaks.